

Yutong Chen

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RESEARCH INTERESTS

Large Language Models, Reinforcement Learning, Data-Efficient Learning

EDUCATION

Clemson University, South Carolina, U.S.
Ph.D. student in Computer Science
CLAIM Lab; Advisor: Kunpeng Liu

Aug 2026 — Present

Tsinghua University, Beijing, China
Master of Electronic and Information Engineering
GPA Rank: 24/99 (24%)
MSIIP Lab; Advisor: Ji Wu

Sep 2023 — Jul 2026
GPA: 3.77/4.00

Tsinghua University, Beijing, China
Bachelor of Electronic Information Science and Technology
GPA Rank: 76/253 (30%)

Sep 2019 — Jul 2023
GPA: 3.77/4.00

RESEARCH EXPERIENCE

CoScale-RL: Efficient Post-Training by Co-Scaling Data and Computation

Yutong Chen, Jiandong Gao, Ji Wu

arXiv: <https://arxiv.org/abs/2601.14695>

Aug 2025 — Jan 2026

Description: Training large reasoning models (LRMs) is often unstable and unpredictable, especially on hard problems or with weak foundation models. We propose CoScale-RL, a novel scaling strategy that improves data and computational efficiency. We first scale up solutions to make problems solvable. The core idea is to collect multiple solutions for each problem rather than simply enlarging the dataset. We then scale up rollout computation to stabilize reinforcement learning. We further leverage a model-merging technique called Re-distillation to sustain or even improve computational efficiency as we scale up. Our method significantly improves data and computational efficiency, achieving an average 3.76-fold improvement in accuracy across four benchmarks. CoScale-RL improves an LRM’s capability boundary without requiring an extensive SFT dataset.

Contribution: This work was done at the Multimedia Signal and Intelligent Information Processing Lab of Tsinghua University under the supervision of Ji Wu and Jiandong Gao. I made major contributions to experiment design, coding, theoretical analysis, and writing.

Investigating the Effectiveness of Small-Scale Fine-Tuning in R1-Style Reinforcement Learning

Yutong Chen, Jiandong Gao, Ji Wu

arXiv: <https://arxiv.org/abs/2505.17988>

Feb 2025 — July 2025

Public Repository: <https://github.com/on1262/deep-reasoning>

Description: Experiments have shown that small-scale supervised fine-tuning (SFT) substantially influences reinforcement learning (RL) but is inefficient. To explain these observations, we propose an analytical framework and compare the efficiency of SFT and RL by measuring **sample effect**. Our theoretical analysis suggests ways to improve SFT efficiency. Guided by this analysis, we propose **Re-distillation**, a technique that aims to boost the effectiveness of small-scale distillation by sampling from an RL-trained policy. Re-distillation demonstrates consistent and substantial efficiency gains across three datasets and both Qwen and Llama models: Re-distilled models matched RL performance with far fewer samples and less computation. On the K&K dataset, our re-distilled Qwen-2.5-1.5B model surpasses DeepSeek-V3-0324 with only 1K SFT samples. We demonstrate that re-distillation can efficiently balance multiple goals in RL. Our work explains several interesting phenomena in R1-style RL, shedding light on the mechanisms behind its empirical success.

Contribution: This work was done at the Multimedia Signal and Intelligent Information Processing Lab of Tsinghua University under the supervision of Ji Wu and Jiandong Gao. I made major contributions to experiment design, coding, theoretical analysis, and writing.

Expressive Speech-Driven Facial Animation with Controllable Emotions

Yutong Chen, Junhong Zhao, Wei-Qiang Zhang

Published in *2023 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*.

Jan 2023 — July 2023

Paper: <https://ieeexplore.ieee.org/document/10223191>

Public Repository: <https://github.com/on1262/facialanimation>

Description: Generating highly realistic facial animation is in high demand but remains challenging. Existing speech-driven facial animation approaches can produce satisfactory mouth movements and lip synchronization but are limited in their ability to generate dramatic emotional expressions and flexibly control emotions. This paper presents a novel deep learning-based approach for generating expressive facial animation from speech, exhibiting a wide spectrum of facial expressions with controllable emotion type and intensity. We propose an emotion controller module to learn the relationship between emotion variations (e.g., type and intensity) and the corresponding facial expression parameters. It enables emotion-controllable facial animation, in which the target expression can be continuously adjusted as desired. Qualitative and quantitative evaluations show that the animation generated by our method exhibits rich facial emotional expressiveness while retaining accurate lip movements, outperforming other state-of-the-art methods.

Contribution: This work was completed during my internship at the Speech and Audio Technology Lab under the supervision of Wei-Qiang Zhang and Junhong Zhao. I was responsible for coding, experiment design, and writing the initial draft.

FELLOWSHIPS

The Robert M. Geist Endowed Fellowship — Aug 2026

ENGLISH & GRE TESTS

TOEFL iBT: 100 (overall score)

Listening: 25 — Reading: 30

Speaking: 20 — Writing: 25

Test date: July 2025

GRE General Test: 324 (overall score)

Quantitative Reasoning: 154 — Verbal Reasoning: 170

Analytical Writing: 3.5

Test date: July 2025

SKILLS

- **AI Infrastructure:** I currently use vLLM and VeRL for development and have experience with OpenRLHF and DeepSpeed.
- **Algorithms:** I am familiar with reinforcement learning for large language models (RLHF, DPO, and GRPO). My knowledge of classical RL includes policy gradient methods (REINFORCE, A2C, TRPO, and PPO) and some value-based methods (Q-learning, DQN, and SARSA).